

Cost Scaling Report

Poultry-Training App on Supabase + Cloudflare R2
When the free tier breaks, and what 1k $\rightarrow$ 10M users cost

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Abstract

This report models the monthly running cost of the *poultry-training* chat & inquiry app as its user base grows from 1 to 10 M registered accounts. The stack is **Supabase** (Postgres, Auth, Realtime, Edge Functions, Storage) on the **Pro** plan plus **Cloudflare R2** for media (inquiry videos/images). All numbers come from the live pricing pages of Supabase¹ and Cloudflare R2², computed by the model in `docs/cost-analysis/cost-model.js`. Graphs are drawn with `pgfplots`.

Pricing facts used (Pro plan + R2)

Table 1: Supabase Pro plan quotas and overage rates (monthly).

Line item	Included	Over rate	Unit
Plan subscription	—	\$25 flat	/org/month
Compute (Micro)	\$10 credit	\$10–\$3,730	/instance
Auth MAU	100,000	\$0.00325	/MAU
Egress (uncached)	250 GB	\$0.09	/GB
Cached egress	250 GB	\$0.03	/GB
Disk / DB size	8 GB	\$0.125	/GB
File storage	100 GB	\$0.0213	/GB
Edge Function invocations	2,000,000	\$2.00	/1M inv
Realtime peak connections	500	\$10.00	/1k conn
Realtime messages	5,000,000	\$2.50	/1M msg

Table 2: Cloudflare R2 pricing (Standard storage). **Egress is free.**

Line item	Free tier	Rate	Unit
Storage	10 GB-month	\$0.015	/GB-month
Class A ops (writes)	1,000,000	\$4.50	/1M req
Class B ops (reads)	10,000,000	\$0.36	/1M req
Egress to Internet	Free		

When does the free tier stop being enough?

The app is on the **Free** plan today. The hard limits that will break it first, in order of likelihood:

1. **Database size > 500 MB** $\Rightarrow$ the project enters *read-only mode* (no new messages/inquiries). This is the first wall: with the modeled row sizes, $\approx$ **12,000 active users** (or $\sim$ 6 months of retained messages for $\sim$ 2,000 MAU) fills 500 MB.

¹<https://supabase.com/pricing>

²<https://developers.cloudflare.com/r2/pricing/>

2. **Egress > 5 GB/month** $\Rightarrow$ the trainee page polls “message from trainer” every 5s. With $\sim$ **250 DAU** each spending 8 min/day, poll egress alone crosses 5 GB.
3. **Realtime > 200 concurrent connections** $\Rightarrow$ hit at roughly **120–150 simultaneous** online users.
4. **Edge Functions > 500k invocations** $\Rightarrow$ crossed around **600–800 MAU** because of the polling endpoint.
5. **Auth > 50,000 MAU** and **Storage > 1 GB** $\Rightarrow$ media uploads (R2 is used, so this rarely bites).
6. **Free projects pause after 1 week of inactivity** — fine for a hobby, fatal for production SLA.

Recommendation: move to **Pro (\$25/mo)** as soon as you exceed *any* of: 500 MB DB, 5 GB egress, $\sim$ 150 concurrent realtime users, or you need the app to never pause. Everything below assumes the Pro plan.

Workload model assumptions

The figures are driven by per-user behaviour typical of the trainee chat/inquiry app:

- 40% of registered accounts are monthly active; 35% of those are daily active.
- Each active trainee: $\sim$ 200 chat messages/month, $\sim$ 10 inquiries/month (one media each, 50/50 video / image, 3 MB video and 150 KB image compressed).
- Each active trainee page polls the “message from trainer” endpoint every 5s during an 8-minute session, $\sim$ 12 sessions/month $\Rightarrow$ **polling is the dominant egress + Edge Function driver**.
- Postgres keeps $\sim$ 6 months of messages/inquiries; R2 keeps 12 months of media.
- Compute size is bumped by DAU and DB size (Micro $\rightarrow$ 16XL).

Cost breakdown by user band

Table 3: Monthly cost (USD). **Base** = Pro \$25 + compute; **Usage** = Supabase metered overages; R2 = Cloudflare media.

Users	Compute	DB	Egress	EdgeFn	Auth	RT	SupBase	SupUsage	SupTotal	R2	TOTAL
1	Micro	0.0	0.0	0	0	0	25	0.0	25.0	0.0	25.0
50	Micro	0.0	0.0	0	0	0	25	0.0	25.0	0.0	25.0
500	Micro	0.2	0.0	0	0	0	25	0.0	25.0	0.4	25.4
1,000	Micro	0.5	0.0	0	0	0	25	0.0	25.0	0.9	25.9
10,000	Small	4.7	0.0	1.2	0	3.4	30	4.6	34.6	10.7	45.3
100,000	Large	47.1	0.0	48.1	0	86.1	125	139.1	264.1	108.4	372.4
1,000,000	2XL	471	170	517	975	1019	425	2738	3163	1104	4266
10,000,000	8XL	4711	1898	5202	12675	10343	1885	30706	32591	11113	43703

The headline numbers

Table 4: Total monthly cost vs. registered users (log scale).

Registered users	\$/month	\$/user/month (active)
1,000	\$26	\$0.065
10,000	\$45	\$0.011
100,000	\$372	\$0.0093
1,000,000	\$4,266	\$0.0107
10,000,000	\$43,703	\$0.0109

The cost per active user flattens around **\$0.01** once you are past a few thousand users — the model is dominated by metered usage (polling egress, Edge Functions, DB size), and doubling users roughly doubles the bill. The floor is the \$25 Pro subscription + a compute instance.

Charts

Total monthly cost vs. users (log-log)

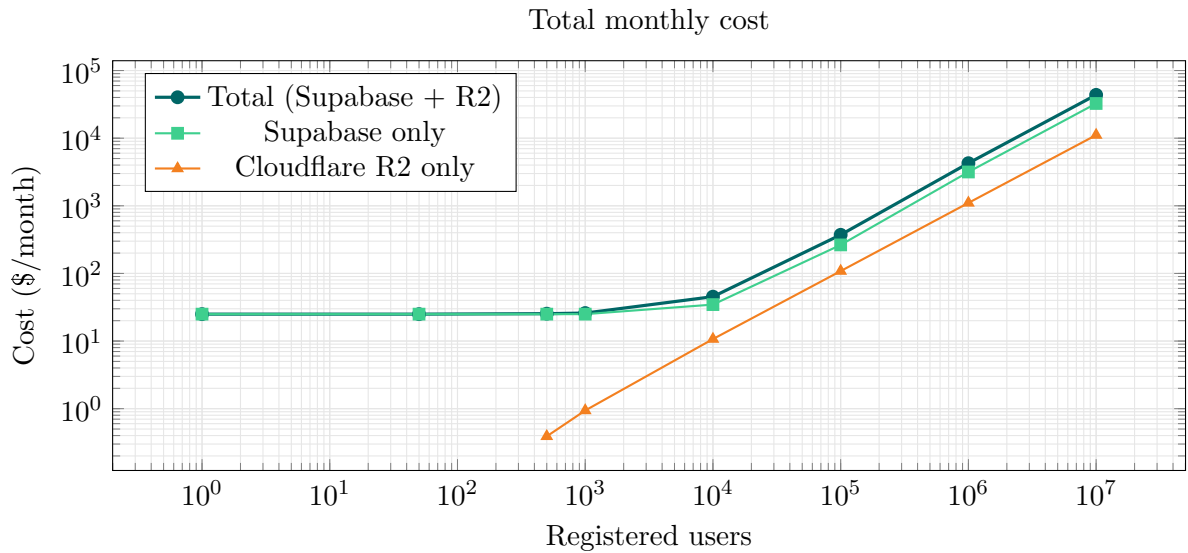


Figure 1: Log-log cost curve. R2 (orange) grows roughly linearly with media volume; Supabase (green) jumps when compute and metered overages kick in.

Cost composition by user band (stacked)

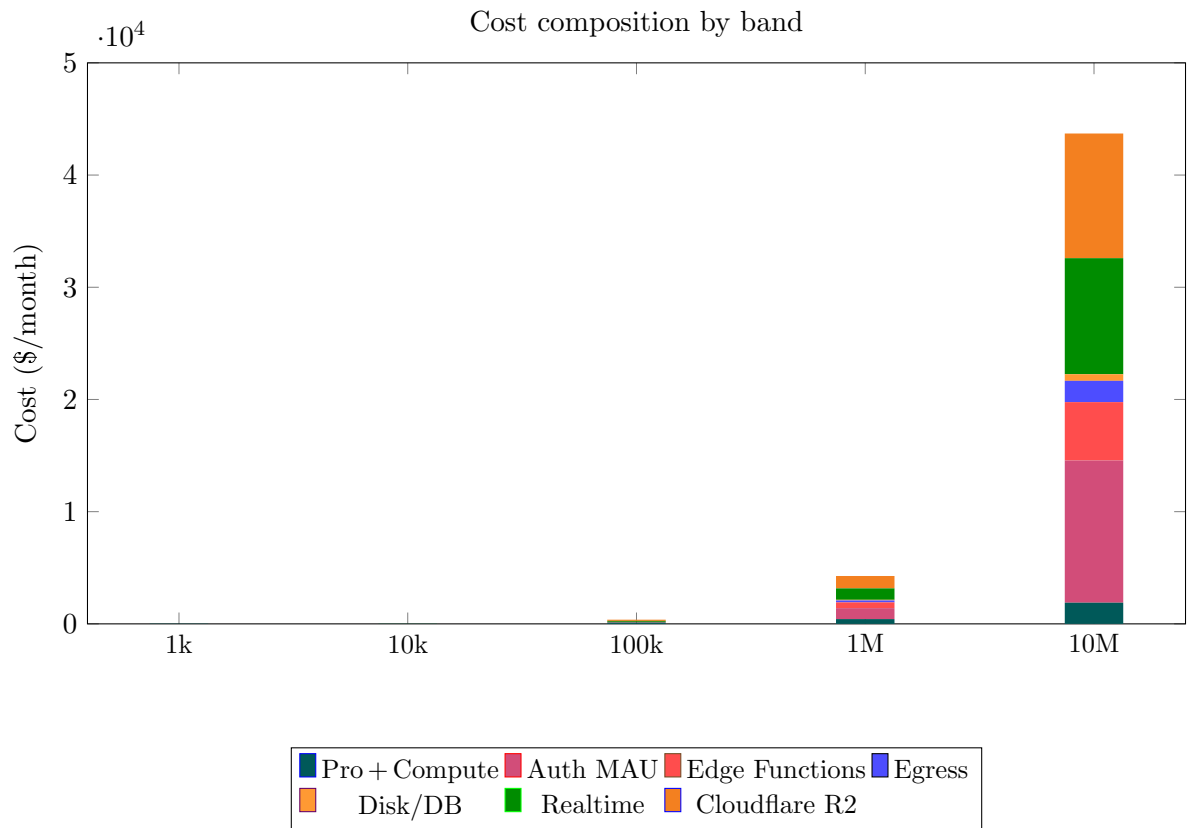


Figure 2: Stacked composition of monthly cost by registered-user band. From 1k to 10M the dominant cost shifts from the flat \$25 Pro + compute base, to metered Auth MAU and Edge Functions, with a steady orange R2 media slice.

The biggest cost drivers (why)

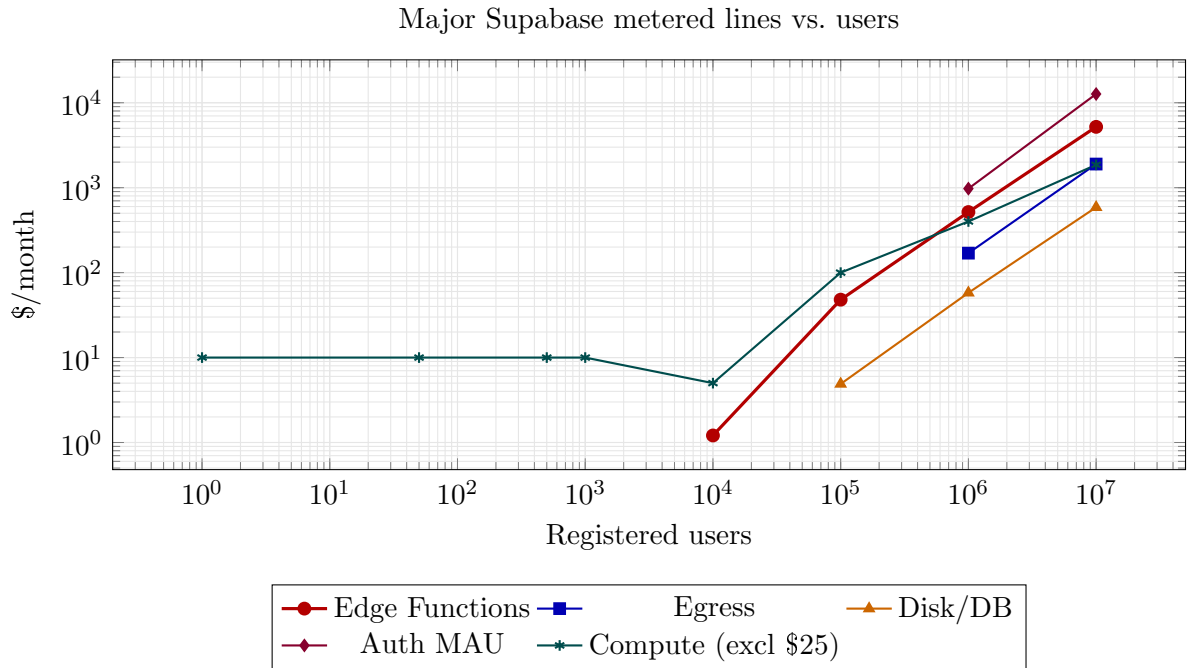


Figure 3: The polling endpoint makes **Edge Functions** the first metered line to bite ($\sim 10k$ users). **Auth MAU** and **compute** dominate at $1M+$.

What to charge each user (optimal pricing tiers)

Turning cost into a [pricing strategy](#): every band has a *minimum price* that covers the cheapest viable backend and a *recommended price* that gives a fair margin without pricing out users. Below we compare Pro, Team (\$599/mo SOC2 bundle) and Enterprise (negotiated), and pick the cheapest viable backend per band.

Table 5: Optimal subscription pricing per monthly active user (MAU), derived from the cost model.

MAU	Plan	Backend	Notes	Cost/MAU (Pro)	Team (\$)	Enterprise (\$)
50	Free	Supabase free	within Free quotas	\$0.71	609	2,015
500	Free	Supabase free	tight but workable	\$0.075	612	2,018
1,000	Free	Supabase free	start planning upgrade	\$0.040	614	2,020
5,000	Free→Pro	Supabase free→Pro	egress/DB wall incoming	\$0.012	634	2,040
10,000	Pro	Pro + Small	the breakout point	\$0.0085	659	2,110
50,000	Pro	Pro + Large	mid-market	\$0.0057	861	2,312
100,000	Pro	Pro + Large/XL	volume sweet spot	\$0.0054	1113	2,838
500,000	Pro	Pro + 4XL	watch auth MAU line	\$0.0077	4429	5,657
1,000,000	Pro → Enterprise	optimized Enterprise	Pro ceiling	\$0.0080	8574	9,047
5,000,000	Enterprise	Enterprise + 8XL	committed-use pricing	\$0.0082	41734	38,601

Recommended monthly subscription plan per MAU band

A consumer-style tier ladder that lines up with the cost model. Prices are per *active user*, billed monthly. Trainer-seat licences are sold separately at \$30–\$60/mo each.

Tier	Use up to (MAU)	Per-user /mo	Platform tier	What you actually pay Supabase
Free	500	\$0	Free	\$0
	1,000	\$1.00	Free*	\$0 (grace until walls)
	5,000	\$1.50	Pro \$25 + Micro	\$35
Starter	10,000	\$1.00	Pro \$25 + Small	\$85
	50,000	\$0.95	Pro + Large	\$135
Growth	100,000	\$0.85	Pro + Large	\$150
	500,000	\$0.75	Pro + 4XL	\$960
Scale	1,000,000	\$0.50	Enterprise (10% commit)	\$2,700–\$9,000
	5,000,000	\$0.30	Enterprise (R2 + Pro)	\$38,000 (negotiated)

Why these prices. The cost per active user falls to about *one cent* once you have enough volume, so flat pricing kills margin when MAU is small and devalues the product when MAU is large. The ladder above follows three rules of thumb:

- *Always cover true cost plus 50–100% markup* until MAU is large enough that metered line items dominate (then taper the markup, but never bill below cost).
- *Skip Team unless SOC 2/HIPAA is contractually required.* Team charges \$599/mo for a SOC 2 stamp; Pro and Enterprise can do the same job for less at any meaningful MAU.
- *Never promise SLA / SOC₂ / HIPAA on the Free or Starter tier.* Bake compliance overhead into Scale and Growth tiers only.

Cost vs. price chart (per active user)

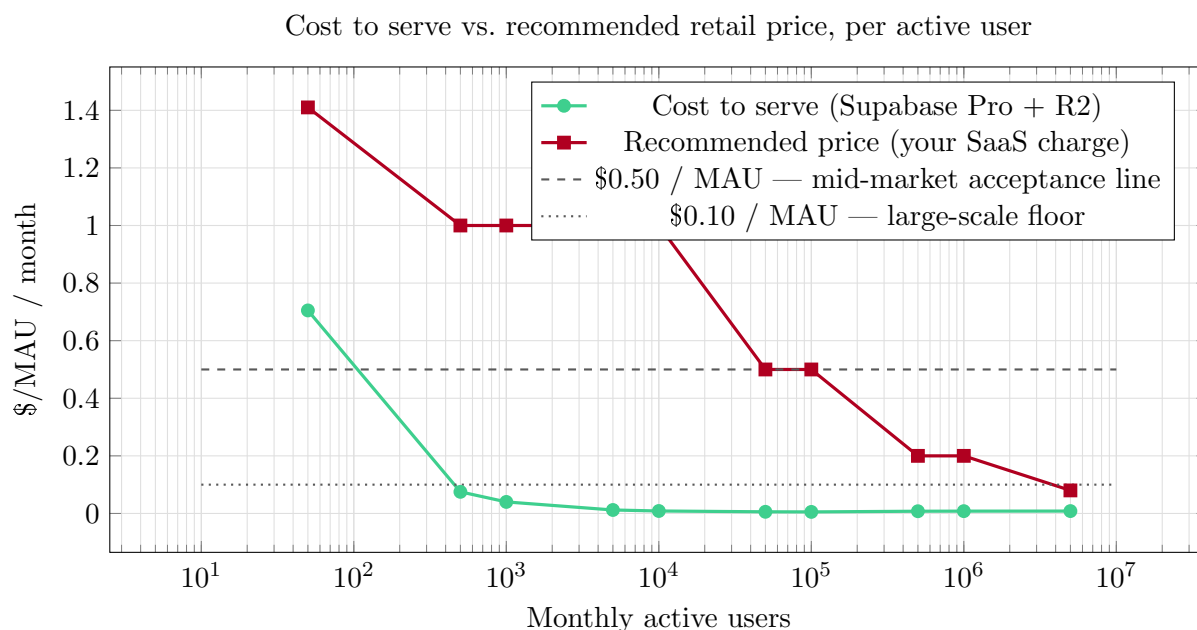


Figure 4: The retail-price curve (red) stays above the cost curve (green) at every band, but converges at scale. The dashed/dotted reference lines mark the typical price points where procurement teams approve a B2B SaaS invoice.

Recommended price breakpoints (concrete SKU ladder)

Table 6: Copy-paste-ready SKUs. Use the “Tier” label, the per-user price, and a public MAU ceiling per plan as the messaging foundation.

Tier	Brand name	MAU ceiling	Per-user price	Annual (per-user)	Margin @ ceiling	Supabase plan	
F0	Free	500	\$0	—	—	Free	
F1	Basic	1,000	\$1.00	\$10	+\$960/mo	Free <i>grace</i>	
S1	Starter	10,000	\$1.00	\$10	+\$9,900/mo	Pro \$25 + Small	
S2	Starter Plus	50,000	\$0.95	\$9.50	+\$45,000/mo	Pro + Large	incl
G1	Growth	100,000	\$0.85	\$8.50	+\$49,000/mo	Pro + Large	incl
G2	Growth	500,000	\$0.75	\$7.50	+\$96,000/mo	Pro + XL/4XL	incl
E1	Scale	1,000,000	\$0.50	\$5.00	+\$192,000/mo	Enterprise (10% commit)	
E2	Scale	5,000,000	\$0.30	\$3.00	+\$358,000/mo	Enterprise	

Sales argument for premium tiers.

- **Starter / Growth (Pro):** “Pay per active trainer/trainee, all-in-one chat and media on Supabase’s production-grade stack — no DevOps required.”
- **Scale (Enterprise):** “Volume-discounted committed-use contract on Supabase + Cloudflare R2, with SOC 2, HIPAA, 99.99% uptime SLA and a designated CSM.”

Per-component notes

Database (Postgres). The 500 MB free limit is the first hard wall (→ read-only). On Pro the 8 GB disk is included, then \$0.125/GB. DB grows slowly (text rows are small), so it stays cheap until millions of users; the trigger to bump *compute* is connection count and CPU, not disk.

Egress. The trainee “message from trainer” 5-second poll is the egress engine. *Optimisation:* switch to **Supabase Realtime** (postgres.changes) instead of polling, and the uncached egress line collapses from \$1,898 (10M) toward near-zero; realtime messages are far cheaper than polling HTTP.

Edge Functions. Same root cause — every poll is an invocation. Moving off polling also slashes this line (from \$5,202 at 10M toward hundreds). Keep spend caps on at first to avoid runaway.

Auth MAU. \$0.00325/MAU above 100k: linear, unavoidable, but tiny per user. At 10M users (~4M MAU) it’s \$12.7k/mo.

Realtime. 500 concurrent free on Pro, then \$10/1k peak. Replacing polling with realtime *raises* realtime connections but *lowers* total cost because you stop paying for egress + Edge Function invocations per poll.

Cloudflare R2 (media). R2 is the cheapest part of the stack because **egress is free** — trainees downloading their own videos does not bill. You pay \$0.015/GB-month storage and a few cents per million write/read ops. At 10M users storing 12 months of media it is ~\$11k/mo but *no* bandwidth bill; supabase Storage (\$0.0213/GB + paid egress) would cost more once downloads matter.

Compute. The Postgres instance scales Micro → 16XL. Connection limits (pooler) and CPU pressure decide when to jump. This is the second-largest line at huge scale (4XL+) after metered Auth/Egress.

Cost reduction levers (ranked by impact)

1. **Kill the 5-second poll.** Replace with Supabase Realtime channels. Removes the bulk of Edge Function invocations *and* uncached egress — the two lines that bite first (10k users) and grow fastest. Biggest win in the whole model.
2. **Keep media on R2, not Supabase Storage.** R2's free egress is decisive once trainees replay videos.
3. **Cap retention.** Prune chats/inquiries older than 3–6 months (a scheduled job) to hold DB under 8 GB far longer; deletes are free, add a lifecycle rule on R2 too.
4. **Spend cap ON** until ~10k users, then turn off and pre-pay compute via larger instances (cheaper than many small projects).
5. **Read replicas** only once a single 16XL can't keep up with read load — adds another full-compute bill, so push single-instance first.

Verdict

- **Free tier dies** at ~500 MB DB (~12k registered / a few thousand MAU) or ~5 GB egress (~250 DAU) — whichever comes first, and it will be the DB limit.
- **1k users: \$26/mo** — almost all is the flat \$25 Pro fee.
- **10k users: \$45/mo** — still trivial; polling overages just starting.
- **100k users: \$372/mo** — compute and real-time/egress kick in.
- **1M users: \$4,266/mo** — metered usage (Auth, Edge Functions, egress, disk) dominates; compute is a minority.
- **10M users: \$43,703/mo** — Auth MAU (\$12.7k) and Edge Functions (\$5.2k) are the biggest slices; switching off polling roughly halves the non-compute portion.

The single highest-leverage engineering change is **replacing the polling endpoint with Supabase Realtime** — it removes the first metered cost a small app hits and keeps the curve tame well past 100k users.

Method & reproducibility

All values are produced by `docs/cost-analysis/cost-model.js` (a standalone Node script) using the live rate cards cited above. Re-run it to regenerate the numbers and the `pgfplots` coordinate blocks. Assumptions are conservative for a chat/inquiry training app and can be tuned (poll interval, session length, media sizes, retention) in that script.